



The University of Texas at Austin
McCombs School of Business

Automated Support-Ticket Triage with a Local LLM

Categorization, tagging, priority and first-response drafting with Mistral-7B-Instruct

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Introduction to Natural Language Processing · Capstone · June 2024

Post Graduate Program in AI & ML: Business Applications

McCombs School of Business, The University of Texas at Austin

PROJECT

06

Introduction to Natural Language Processing · Capstone

A four-stage prompting pipeline on Mistral-7B-Instruct, run locally with llama.cpp, that categorizes, tags, prioritizes and drafts replies to customer support tickets as structured JSON.

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Executive summary

4

chained prompting tasks: category
→ tags → priority and ETA → draft
reply

21

tickets processed end to end into
structured, valid JSON

7B

parameter open model (Mistral-7B-
Instruct v0.2, Q6_K), run locally —
no data leaves the machine

~12 min

total inference for all four tasks on
21 tickets, temperature 0.01

Key findings

Prompt design carries the system. Explicit JSON schemas, few-shot examples and guideline rules produced parseable output on every ticket; each task feeds the next as context.

The model over-assigns urgency. 15 of 21 tickets came back High priority despite an instruction to balance the distribution — customers write urgently, and the model follows.

Hardware dominates the queue. 8 of 21 tickets; data recovery and internet issues 4 each. ETAs cluster at four hours.

Business problem and solution approach

Problem

- Support tickets arrive as free text. Someone has to read each one, decide what it is about, how urgent it is, when it can be resolved, and write a first reply. That triage is slow, inconsistent and expensive at volume.

Objective: use a large language model to perform all four steps automatically and return structured fields a ticketing system can consume.

Approach

- Load a quantized open model locally with llama.cpp (privacy; no API cost)
- Design one prompt per task with a strict JSON output contract
- Chain the tasks: each stage receives the previous stage's output as context
- Parse and normalize responses into a single table
- Analyze the distribution of categories, priorities and ETAs; derive recommendations

Data and model

Data

21 support tickets, two columns: ticket ID and free-text description. No missing values, no duplicates.

Typical ticket: "Urgent help required! My laptop refuses to start, and I have a crucial presentation scheduled for tomorrow..."

No labels. There is no ground truth; evaluation is by inspection of output validity and plausibility.

Model and runtime

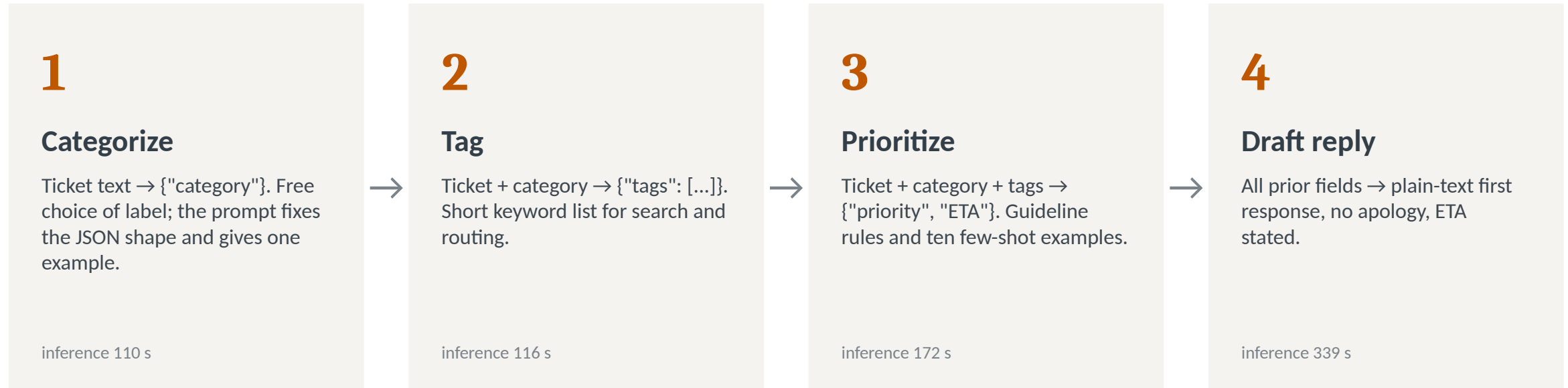
Mistral-7B-Instruct v0.2, GGUF Q6_K quantization from TheBloke, downloaded from the Hugging Face Hub.

llama-cpp-python 0.1.85, context window 1,024 tokens, GPU offload where available (Colab T4).

Decoding: temperature 0.01 for near-deterministic output; max 1,024 tokens; echo off.

Parsing: output sliced from the first '{' and loaded as JSON; lists normalized into columns.

The four-task pipeline



Chaining matters. Each stage is conditioned on the structured output of the ones before it, so the priority prompt sees the category and tags rather than re-deriving them, and the reply prompt can state the ETA it was given.

Output contract. Every prompt specifies the exact JSON keys and an example; the parser takes the substring from the first brace. All 21×3 structured responses parsed without repair.

Prompt 3 — priority and ETA: rules and examples

The most engineered of the four prompts. Priority is a judgment call, so the prompt constrains it with rules, examples and an explicit request for balance — which the model only partly honored.

Element	Content
Role	"You are an AI tasked with assigning priority levels and estimated time of arrival (ETA) to support tickets."
Schema	{ "priority": "priority_level", "ETA": "estimated_time_of_arrival" }
Guidelines	High → significant disruption or urgent need, ETA a few hours · Medium → impactful but non-critical, ETA a day · Low → minor, general questions, ETA three days
Rule overrides	Tags containing optimization, battery, gaming or touchpad → Low
Balance instruction	"Aim for a balanced distribution of priorities."
Few-shot	Ten ticket → JSON examples spanning High (2 h), Medium (1 day) and Low (3 days)

Worked example — ticket ST2023-008

Stage	Output
Ticket	"I've accidentally deleted essential work documents, causing substantial data loss. I understand the need to avoid further actions on my device. Can you please prioritize the data recovery process and guide me through it?"
1 • Category	{ "category": "Data Recovery" }
2 • Tags	{ "tags": ["data recovery"] }
3 • Priority / ETA	{ "priority": "High", "ETA": "4 hours" }
4 • Draft reply	"We understand the urgency of your issue regarding the deletion of essential work documents and the significant data loss. Our team will prioritize the data recovery process and get back to you within 4 hours. In the meantime, please avoid using your device to prevent any further data loss."

WORKED EXAMPLE

Worked example — three more tickets

Categories are specific and sensible; tags are short and searchable; ETAs scale with the model's reading of urgency.

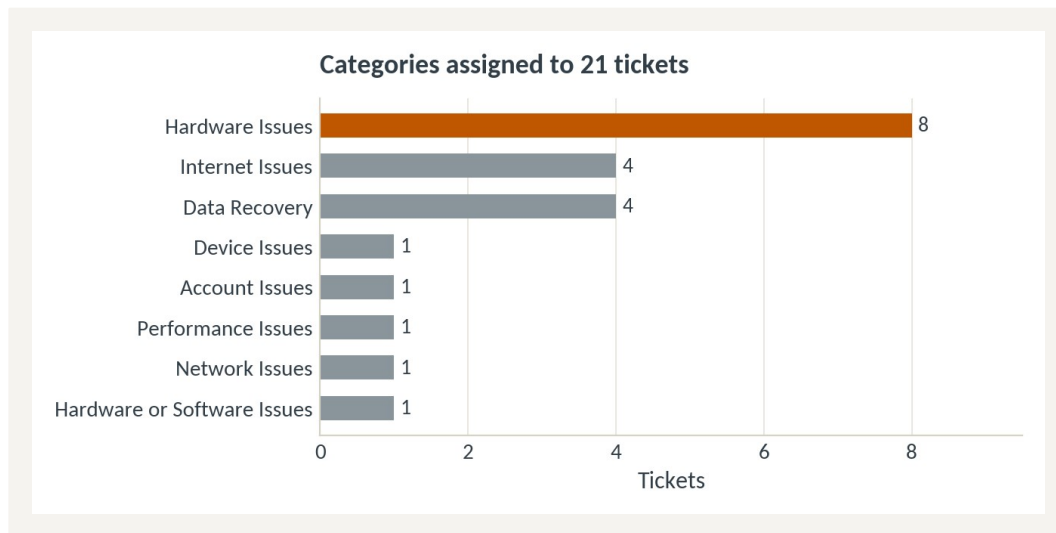
Ticket	Category	Tags	Priority	ETA
ST2023-007 "Urgent help required! My laptop refuses to start, and I have a crucial presentation scheduled for tomorrow..."	Hardware Issues	hardware, laptop, startup	High	2 hours
ST2023-006 Internet connection has slowed significantly; video calls dropping	Internet Issues	internet, connectivity, slow connection, disconnection	High	4 hours
ST2023-011 Locked out of online banking account	Account Issues	account, password, banking	High	1 hour

Model output analysis

What the model assigned

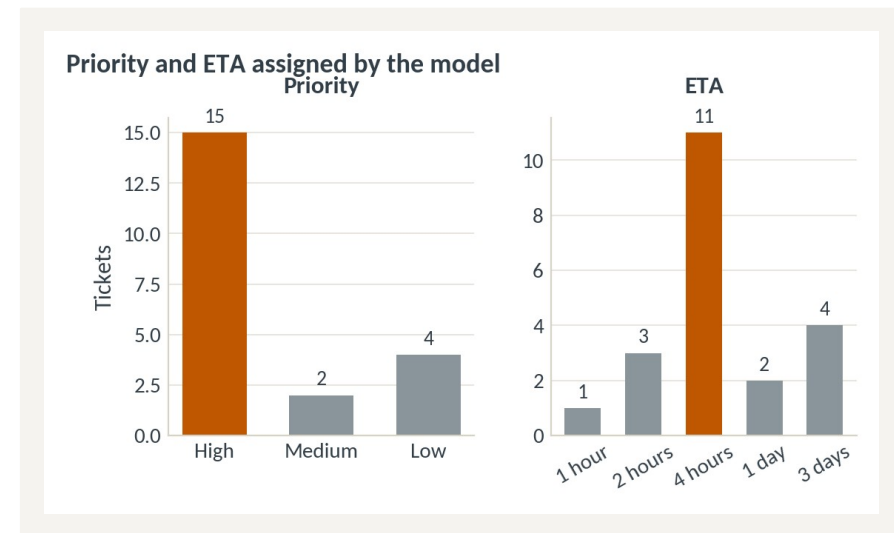
Category

- Hardware issues lead (8 of 21); data recovery and internet issues follow (4 each). Five categories were used once — including 'Hardware or Software Issues', a hedge the prompt did not forbid.

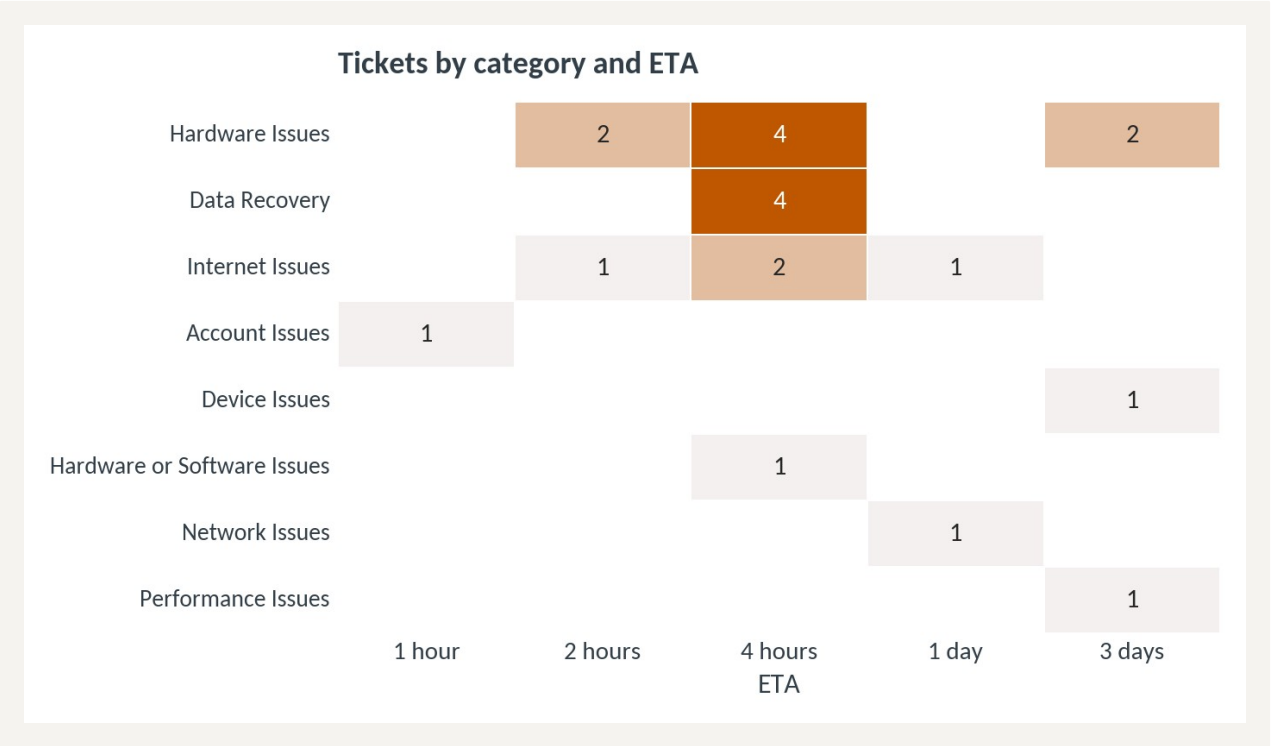


Priority and ETA

- Fifteen High, four Low, two Medium. ETAs cluster at four hours (11); three days for the Low tier (4).



Category against ETA



Data recovery → 4 hours, every time. Consistent, and appropriate for a critical issue.

Hardware issues spread across 2 hours, 4 hours and 3 days, reflecting different severities within the category.

Internet issues: 1 day, 2 hours and 4 hours — connectivity problems read as needing a fast turnaround.

Account issues → 1 hour: the fastest ETA, for a lockout that blocks the customer entirely.

Insights and recommendations

01

Strengthen hardware support

Hardware is the largest category. Allocate resources or a specialist team to it.

02

Improve data-recovery process

Four tickets, all High, all four-hour ETAs. Tools and training to shorten recovery.

03

Optimize connectivity support

Frequent internet issues call for better network tooling and proactive monitoring.

04

Triage the High tier

Fifteen High tickets means the label no longer discriminates. A triage step inside the High tier — or a stricter prompt — is needed.

05

Standardize ETAs

Similar categories receive a wide range of ETAs. Review and standardize where appropriate so customers get consistent expectations.

06

Mine the tags

Use tag frequencies to spot recurring problems and streamline their resolution paths.

Limits and next steps

No ground truth. With 21 unlabeled tickets, the evaluation is qualitative. A labeled set of a few hundred tickets would allow accuracy, agreement with human triage and calibration of ETAs to be measured.

Priority skew. The balance instruction and rule overrides did not stop 71% of tickets landing in High. Options: a stricter rubric, a fixed quota, or a second pass that re-ranks within the High tier.

Category drift. Free-text categories produced near-duplicates (Device / Hardware / 'Hardware or Software'). A closed label list in the prompt would fix this.

Latency. Twelve minutes for 21 tickets is fine for a batch job, not for live triage; a smaller quantization or batching would be needed at volume.



Thank you

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